

Course plan

1. homework from last week (CS! shape)

refer to the instructions from last week

2. Topic for this week

Basic intro to variables in C++ and how to read data into variables

2.1 Type of variables

int: any integer (whole number) like 11, 43, -25 (around 2 billion)

long long: large integers

char: any single character like 'a', '%', 'M'

string: a sequence of characters like "paul cao"

bool: true or false

double: any decimal number with very high precisions like 3.1415926

float: any decimal number with high precisions like 2.25, -1.5

Exercise: Declare variables to store the following data

- age of a person
- day of a week such as sunday
- month as a value such as 3, 12
- a character
- numerical value for pi
- money amount in whole dollars
- last name of a person
- how much money Jeff Beso has in whole dollars
- the height of a person
- the weight of a person
- the number of students in our class
- the average of 8 numbers

2.2 Exercise

Write a code that convert miles to kilometers. The user will input a distance in miles and the program will calculate the corresponding distance in kilometers.

3. Homework for this week

In the Paris Olympics games in 2024, the high jump gold medal for men has a height of 2.36 meters. How many feet is that? Please complete the following code for converting meters to feet. You will read in a decimal value for meter and you should output its equivalent feet amount.